Homework Set 3C

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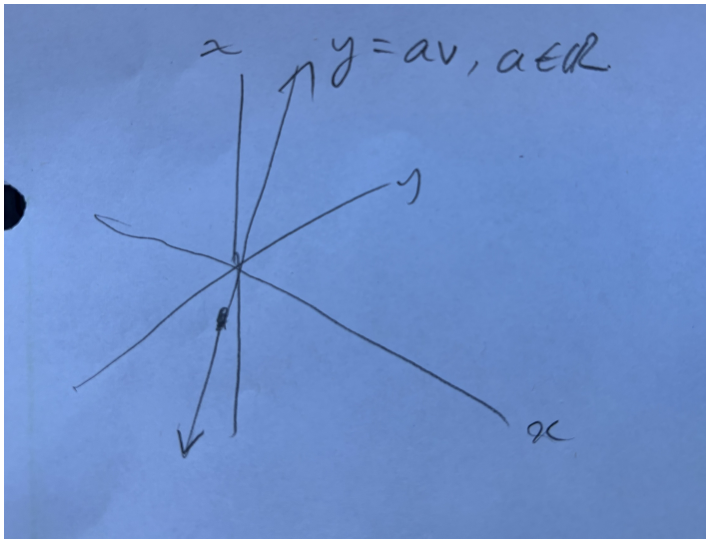
September 11, 2025

Problem 1

Any point (x, y, z) on the line is a scalar multiple of the vector $v = (1, -2, 3)$:

$$(x, y, z) = av = (a, -2a, 3a) \iff \begin{cases} x = a \\ y = -2a = -2x \\ z = 3a = 3x \end{cases}$$

A point is on the line if and only if we can write it in the form $(x, -2x, 3x)$, where $x \in \mathbb{R}$.



Problem 2

Any satisfying the equation $y = 2x$ can be written as ($a curve or other figure formed by all the points satisfying a par$ where $x, z \in \mathbb{R}$. Since there are 2 free variables, the set of points satisfying the equation is a plane.

